

Computing Project - Final Report

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1 Introduction

The purpose of this project is to create a model which can accurately determine the temperature gradient within a steak over time. This model will be able to provide insight into the cooking process with variable oven temperature over time. This model will then be used to optimize the cooking strategy of the steak to produce the desired result of creating a thin char layer and as uniform of internal temperature as possible without exceeding medium rare temperatures. The final task will be to use the model to be able to predict the time until the steak is cooked based on the temperature history of the steak with the assumption that not all system parameters are known.

2 Task 1 - Numerical Model

2.1 Procedure

Colab Link

To begin this task we followed the general outline provided for solving the finite volume method in cylindrical geometry. We created vectors for the radial and vertical directions which would come together to form our mesh. We also created a time vector and an initial temperature matrix, which was the same size as the mesh. We had to initialize the temperature matrix over the entire mesh to show that the initial temperature of the steak was consistent throughout. We defined a function called `TinfFunc` which would take in a time and output a temperature. This would allow the user to have a time variable oven temperature, however for our purposes we keep the temperature constant since this was easier for use to visually validate the temperature gradient. Our main function takes in both a time and a flattened version of the temperature matrix. We had to flatten the temperature matrix since we wanted to use the `solve_ivp` method and this could not handle the 2D temperature matrix. Within our main function we first reshape the flattened temperature matrix and

create an initialized dTdt matrix of the same shape. We then looped through both the radial and vertical directions with two nested for loops. Within the for loops we defined the control volume via Equation 1.

$$V_{i,j} = 2\pi r_i \Delta r_i \Delta z_j \quad (1)$$

After that we determined the radial fluxes which would be coming into and out of our control volume via Equation 2.

$$q_{r,i+1/2,j} = -2\pi r_{i+1/2} k \frac{T_{i+1,j} - T_{i,j}}{r_{i+1} - r_i} \quad (2)$$

This equation shows the radial flux on the right side of the control volume and so the radial indices were shifted left by one in order to calculate the radial flux on the left side of the control volume. In a similar manner, the axial flux was calculated, with Equation 3 for the upper side of the control volume.

$$q_{z,i,j+1/2} = -2\pi(r_{i+1/2}^2 - r_{i-1/2}^2)k \frac{T_{i,j+1} - T_{i,j}}{z_{j+1} - z_j} \quad (3)$$

To calculate the lower axial flux, the axial indices were shifted down by one. It was necessary to also include conditional statements in order to impose our boundary conditions when we reached the boundaries. We imposed four boundary conditions, two of which represented the convection at the exterior surfaces and the other two represented the symmetric boundary condition which occurred at the central axis where the radius was zero and at the "bottom" surface of the steak when the height was zero. While the central axis being symmetric was intuitive, a good way to physically think about the symmetric boundary for the "bottom" of the steak is to think of it physically as the middle of the steak, so there is an identical half below which gives the symmetric boundary condition. You could also consider it as being insulated on the bottom but this does not practically make sense. After imposing the proper boundary conditions via conditionals, the net flux across the control volume was calculated with Equation 4.

$$q_{net} = -q_{r,i+1/2,j} + q_{r,i-1/2,j} - q_{z,i,j+1/2} + q_{z,i,j-1/2} \quad (4)$$

This allowed for the time derivative of the temperature to be computed across the entire mesh as Equation 5 shows.

$$\frac{dT}{dt} = \frac{q_{net}}{\rho c_p V_{i,j}} \quad (5)$$

All of this set up was done to spatially discretize the problem which left us with an initial value problem. As mentioned before, we used the `solve_ivp` function with the BDF method. We chose the BDF method since it was an implicit method and we wanted to ensure that our solution would be unconditionally stable as we were unsure if our time step was small enough for an explicit method to be stable. The final part of our code was graphing the results. We wanted to have a visualization for many different variable relationships over time but we found that the most helpful were the heat maps which plotted the temperature over the mesh for a specified time. These results can be seen in the next section.

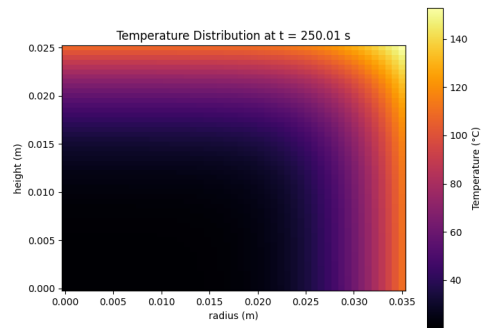
2.2 Results

The desired results for this first task were to create a model which could take in a user defined temperature function and output multiple heat maps of various times as well as how long it would take for all of the steak to reach at least the medium rare temperature of 55 degrees Celsius. Table 1 compiles the project parameters used for producing the following results, as shown in the table we defined a constant oven temperature for easy visual validation but the code is capable of a time variable oven temperature. Below are also the heat maps for the steak for various times throughout the cooking process as well as the total time it took the steak to reach medium rare which was 1515.4 seconds which is just under 25 and a half minutes.

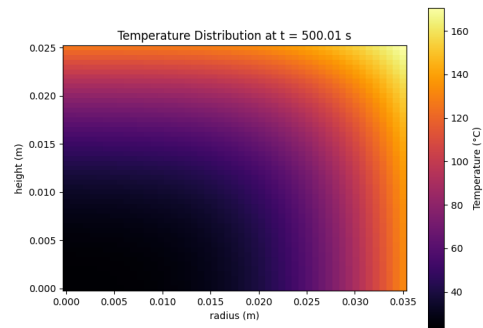
Table 1: Project Parameters

Variable	Name	Value	Units
k	Thermal Conductivity	0.45	W/m/K
ρ	Density	1050	kg/m ³
c_p	Specific Heat	3500	J/kg/K
h	Convection Coefficient	20	W/m ² /K
R	Radius	0.035	m
l	Height	0.025	m
T_{init}	Initial Temperature	293.15	K
T_{∞}	Oven Temperature	473.15	K

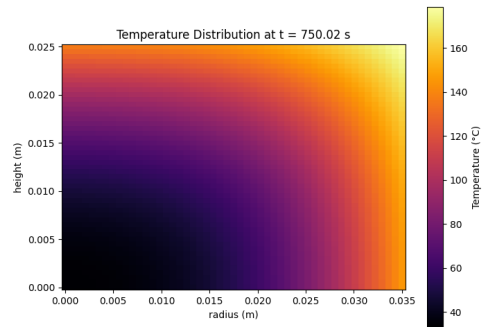
2.2.1 Project Conditions



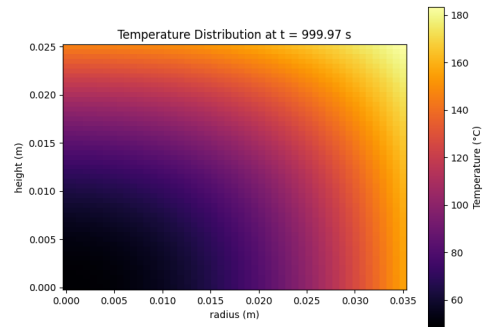
(a) Time: 250 s



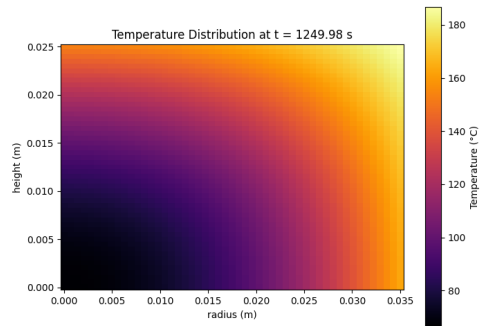
(b) Time: 500 s



(c) Time: 750 s



(d) Time: 1000 s



(e) Time: 1250 s

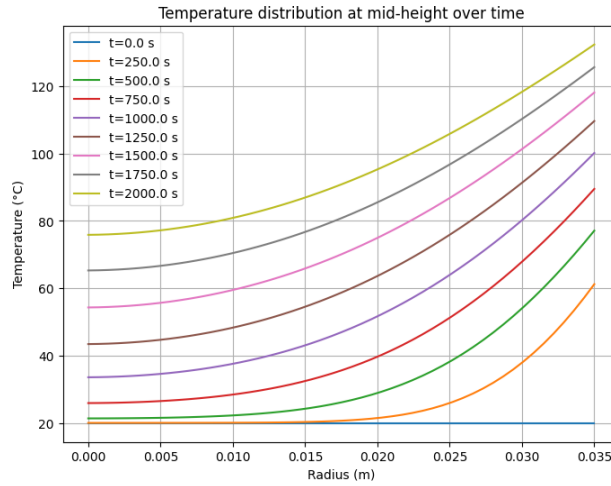


Figure 1: Temperature distribution at different times, at $z = 0$

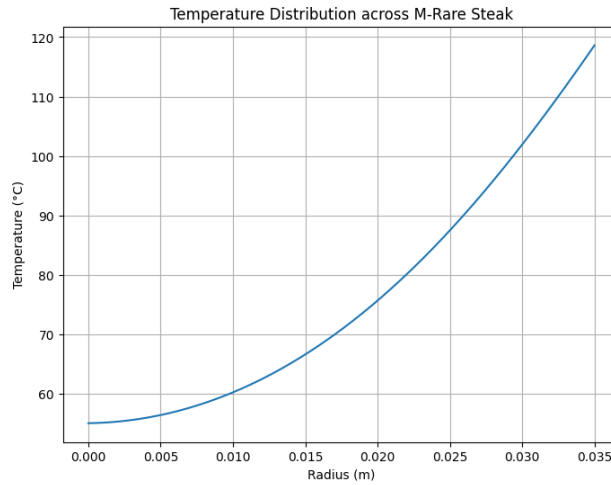


Figure 2: Temperature across the radius when medium rare is achieved

Total time to cook M-Rare = 1515.4 s

2.3 Verification

The cooking project Task 1 code was tested by comparing it to a known solution from Incopera and Dewitt [?]. The problem entails the cooling of a cylinder in an oil bath. The parameters of the problem can be found in Table 2 below:

Table 2: Problem parameters for 2D transient convection cooling for cylinder

Variable	Name	Value	Units
k	Thermal Conductivity	17.4	W/m/K
ρ	Density	7900	kg/m ³
c_p	Specific Heat	526	J/kg/K
h	Convection Coefficient	500	W/m ² /K
R	Radius	0.04	m
l	Height	0.03	m
T_{init}	Initial Temperature	600	K
t_f	Total Simulation time	180	s

The full problem can be found in Appendix A. Using these parameters, our code was used to simulate the transient heat loss in the cylinder. The results of this simulation can be found below in Table 3:

Table 3: Simulation results for temperature distribution in steak cylinder

Location	Simulated T (K)	Expected T (K)	Approx. Error
Center of cylinder	406.50	405	0.37%
Top outer ring of cylinder	344.63	344	0.18%
Center outer ring of cylinder	365.08	365	0.027%
Top center of cylinder	373.05	372	0.287%

Refer to Appendix B to view enlarged heat map figures. As seen from the approximate error data in Table 3, our model accurately predicts the desired temperature distribution, thus validating our code.

3 Task 2 - Optimization

3.1 Procedure

With our program working, we set out to optimize control of the oven temperatures to cook a steak to medium-rare internally while also obtaining a 0.25 cm char layer on the outside. To obey the laws of thermodynamics, we used a conventional apartment oven, focusing on the center temperature. For a conventional home oven, the minimum temperature is 93.33 °C, with a maximum temperature around 288 °C.

In order to optimize the numerical analysis, an objective function was created. This objective function takes the numerical solution to create temperature, char and uniform errors across the solution. The temperature error is defined as:

$$T_{\text{error}} = (T_{\text{center}} - T_{\text{mrare}})^2 \quad (6)$$

Using the difference of squares from the temperature of the center from the numerical solution and the goal T_{mrare} , we can define an error variable which is used in the minimization. Next, we define an error for the char region, defining it:

$$\text{char}_{\text{error}} = (\text{char}_{\text{thickness}} - 0.0025)^2. \quad (7)$$

Similar, to the temperature error, when defining the char error we use the current char depth and the maximum char. Lastly, we define an error to keep the center close to medium rare. The uniform error can be defined in a slightly more complex manner

$$\text{mask} = (rr < R - 0.0025) \ \& \ (zz > 0.0025) \ \& \ (zz < Z - 0.0025) \quad (8)$$

$$\text{vol_error} = \sqrt{\frac{1}{N} \sum_{i \in \text{mask}} (T_{\text{field},i} - T_{\text{mrare}})^2} \quad (9)$$

First, a boolean mask is created the same size of the temperature output, this identifies all non-non charred regions of the 2D domain. The second line computed the root mean square error (RMSE) between the simulated temperature field and the reference temperature T_{mrare} .

Once these three errors are defined, they were weighted based on relative importance. In this case, the center temperature was weighted most important, with the uniform temperature gradient following and the char layer being the least important. The char layer was determined to be the least weighted because of the conditions of the cooking vessel. If we were able to use a hotter cooking element, or introduce conduction then it would be easier to create a char layer without overcooking the steak. With the current conditions of the oven, and steak size it is impossible to cook a steak and achieve a char layer before the center is overcooked.

3.2 Results

By utilizing `scipy.optimize`, we were able to find the optimal cook time and temperature distribution for a cylindrical steak as outlined in Task 1. The results of the optimization can be found in Table 4.

Table 4: hello

$T_{\infty,1}$	$T_{\infty,2}$	t_{switch}	t_{total}
140.65 °C 285.17 °F	224.03 °C 435.25 °F	196.87 s	1971.54 s

After conducting the numerical analysis utilizing these inputs, a heat map could be generated to show the temperature distribution across the steak. Figure 3 shows the temperature gradient across the steak using the numerical solver.

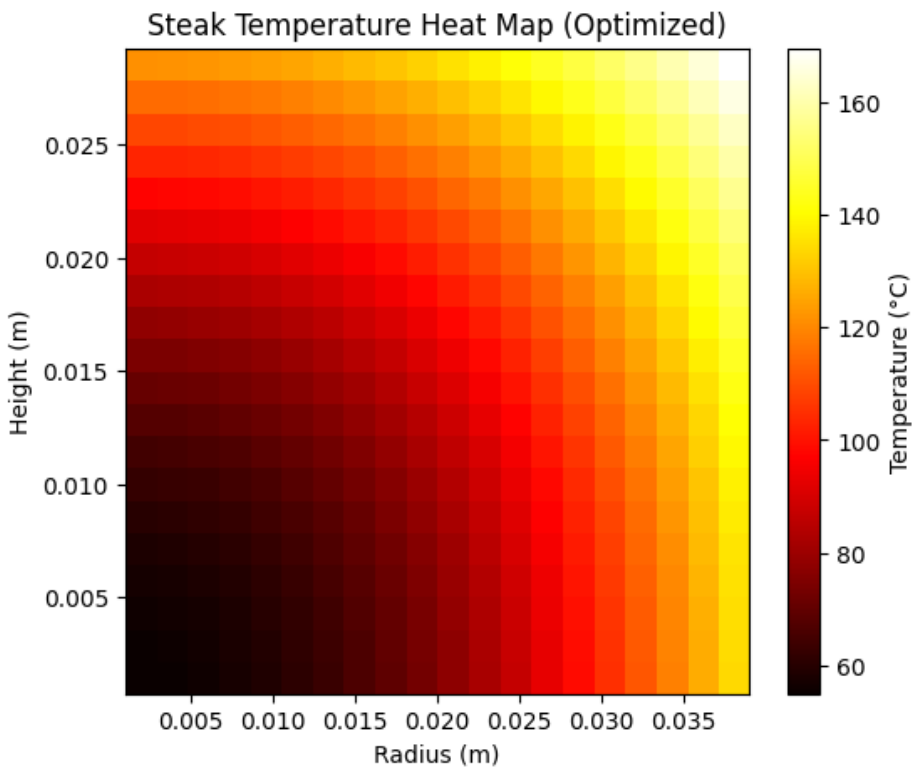


Figure 3: Optimized temperature gradient of steak

The analysis done in Task 2 shows that with a conventional oven it is impossible to get a proper char layer while optimizing the internal temperature of a steak. This analysis has a few major flaws, conduction would need to be introduced into the problem and or a larger steak would need to be analyzed. If conduction were utilized as the boundary conditions, then the char layer would be easier to achieve as the temperature at the boundary layer is not dependent on the convection in the oven, but rather the amount of contact the steak has with a pan. The diffusion of thermal energy is much greater because of conduction than because of convective effects. As for the size of a steak, because we are only analyzing steaks around 4 (in) in diameter as the internal temperature of the steak is raised due to Newton’s

law of cooling, the external temperature does not exceed the char temperature by the time the steak reaches medium rare.

4 Task 3 - Cooking Predictions via Algorithm

4.1 Procedure

For task three, our goal was to teach a machine learning model which was capable of estimating the remaining cooking time for a steak based on a few easily identifiable system variables. These included the time the steak had already been in the oven, the temperature of the oven, which we assumed would be constant, and then four temperature measurements of the steak. We chose to use temperature measurements at the four corners of the heat maps, which represented the center of the steak, the top center of the steak, the outer center of the steak and the top outer of the steak. We wanted the model to be able to use this limited information to predict the cooking time because this mirrors more reasonable real life situations where one does not know the specific material properties of a steak, thus the convection coefficient h was assumed to be unknown.

The first step of creating our model was to generate a data set. Since the main unknown was the convection coefficient, we ran multiple simulations for various values to generate a range of data. Within the variation of the convection coefficient, we also varied the oven temperature to increase the scope of our data set. We ran through 11 different h values ranging from 20 to 70 and 24 different oven temperature values ranging from 120 to 350 degrees Celsius. From each of these initial conditions we saved values for the temperature at each of the four corners of the heat maps as well as the time until the center would reach medium rare. We compiled all of this into a large data set with 40744 samples, which we wrote to a .csv file in order to be imported to train our model.

We decided to use a random forest model since it was the one we felt most confident about implementing when we learned it in lecture. We followed the standard procedure of splitting the data into 80 percent train data and 20 percent test data. The input data set X , contained the time the steak had been in oven, the oven temperature and our four corner temperatures of the heat map. The output data set y , was the time remaining until the middle of the steak would reach medium rare or 55 degrees Celsius. We set the number of estimators to 1000 since this was adequately large to produce a good model but also not too large to cause significant computational delay.

The last step was implementing the model with a user interface to predict the necessary cook time until the center reached medium rare. The user can input the time the steak has been in the oven, the temperature of the oven and the temperature values for the four corners of the temperature map. The model then predicts the time it will take for the center of the steak to reach medium rare and the user will see both how much longer it will take

the steak to be cooked as well as the total expected cooking time.

4.2 Results

After collecting data, the information was split so 80% went into training the model while the other 20% was used to test our algorithm. Figure 4 below shows the importance of each variable to the prediction that our model makes.

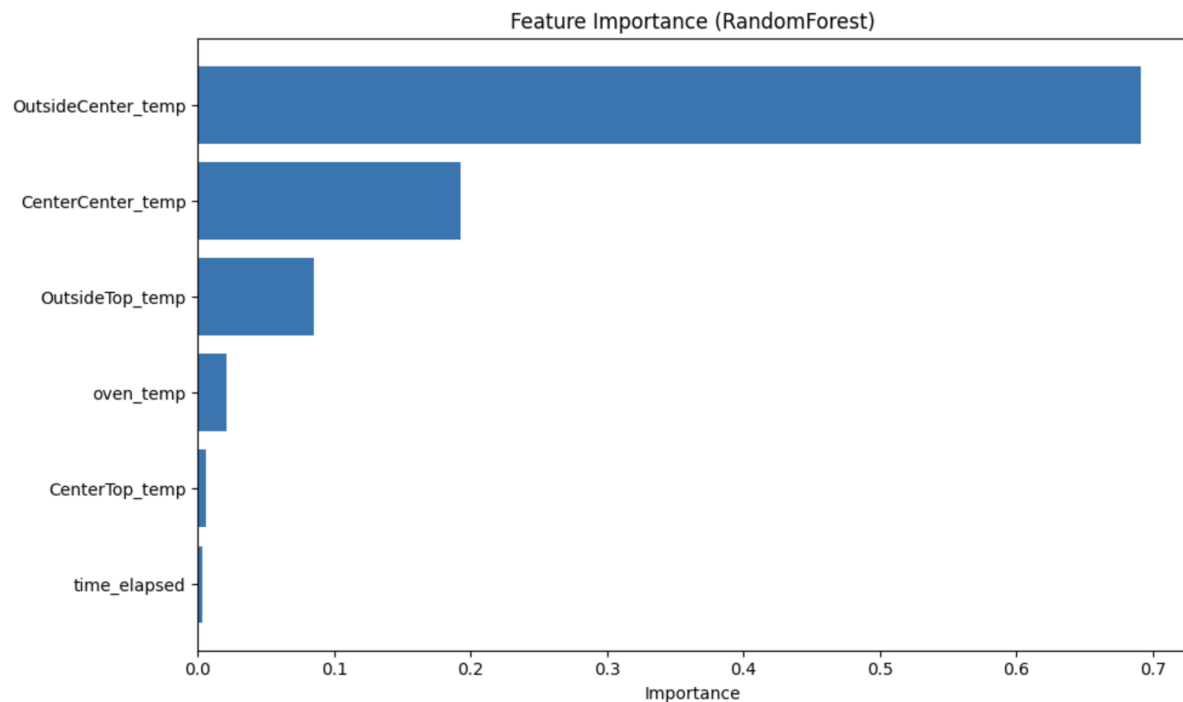


Figure 4: Importance of each feature to the model's predictions.

The most important feature is OutsideCenter_temp because there is more surface area due to the radial definition of the cylinder and thus holds the most weight as the algorithm predicts what temperature and time remaining is needed for each steak.

When using the data set aside for testing, our model performed extremely well. With 1000 estimators, our model had a mean absolute error from actual cooking time to predicted cooking time of 0.01 seconds, and a 90th percentile of absolute error of 0.02 seconds. The models prediction alongside the actual data can be seen in Figure 5 below.

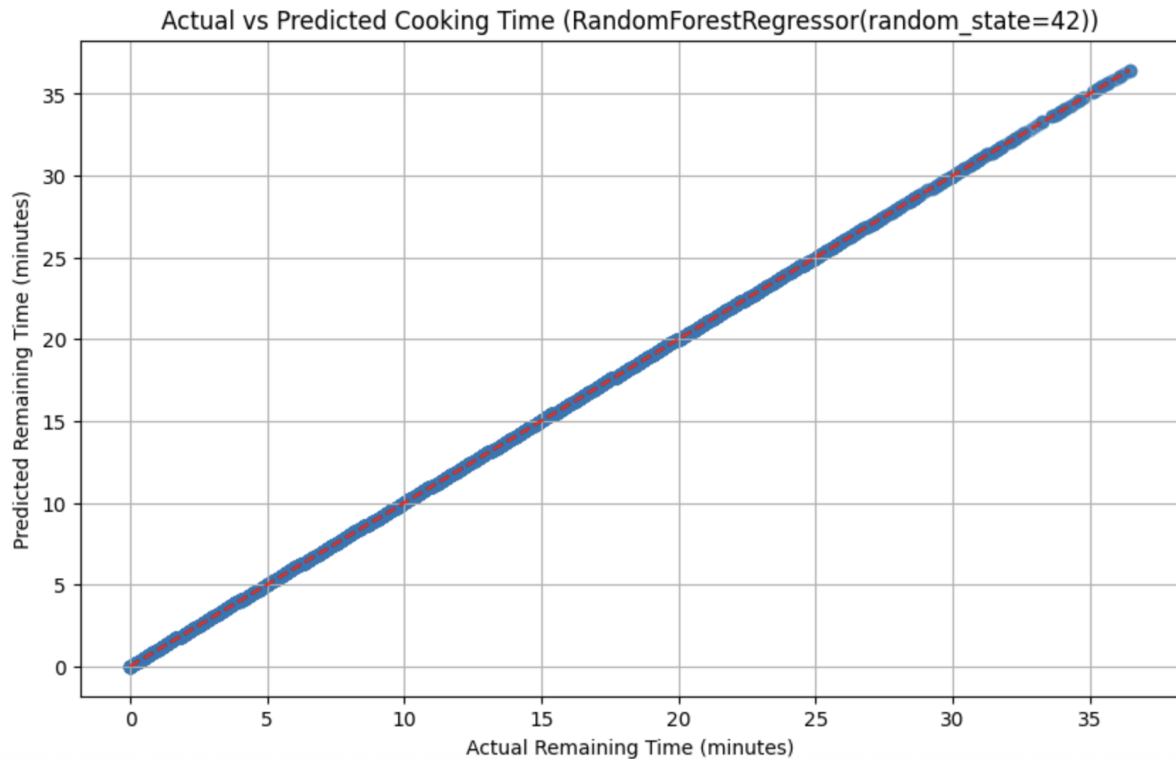


Figure 5: Actual cooking time versus predicted cooking time.

As seen from our absolute error as well as Figure 5, our model's predictions are spot on with what the expected cooking time is.

5 Conclusion

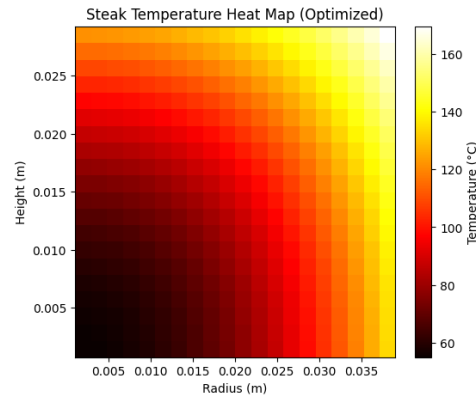
After completing this project we decided to try our models accuracy by cooking a real steak. There were a number of assumptions that we made when creating this model that we knew would cause some initial discrepancies between the real cooking outcome, however it would be a good experiment to see how close the model was to predicting real world cook times. We knew that our steak was not a perfect cylinder and that it had far from a uniform makeup. We also knew that some of our model constants such as density and convection coefficient would not be as accurate. All of these imperfect assumptions caused error between our model and our actual cooking time. However, we were able to test both our optimization and our machine learning model against real data. For our optimization, we followed the suggested cooking temperatures and time. First cooking at 140 degrees Celsius (284 degrees Fahrenheit) for 194 seconds, then we turned up the heat to 224 degrees Celsius (435 degrees Fahrenheit) for 1775 seconds which was about 30 minutes. Unfortunately, we found that this was not long enough and we had to leave the steak in the oven for around another 20 minutes. We had also tested the machine learning model shortly after we had increased the

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temperature of the oven. Unfortunately, this model also underestimated how long the steak would actually need to cook. We believe that these underestimations are due to inaccurate system parameters. You can see a comparison between our actual cooked steak and the heat map generated from our optimization code below.



(a) Real Steak



(b) Modeled Quartered Steak

Like most things in life, cooking a steak never happens under perfect conditions. Whether its the variability of the oven temperature or the not perfectly cylindrical steaks being cooked, there are some discrepancies between our model's predictions and the actual time it takes to cook a particular steak.

Appendix A - Verification Problem

where $P(x, t)$ and $C(r, t)$ are defined by Equations 5S.2 and 5S.3, respectively. Accordingly, for the center of the cylinder,

$$\frac{T(0, 0, 3 \text{ min}) - T_{\infty}}{T_i - T_{\infty}} = \frac{T(0, 3 \text{ min}) - T_{\infty}}{T_i - T_{\infty}} \bigg|_{\text{Plane wall}} \cdot \frac{T(0, 3 \text{ min}) - T_{\infty}}{T_i - T_{\infty}} \bigg|_{\text{Infinite cylinder}}$$

Hence, for the plane wall, with

$$Bi^{-1} = \frac{k}{hL} = \frac{17.4 \text{ W/m} \cdot \text{K}}{500 \text{ W/m}^2 \cdot \text{K} \times 0.03 \text{ m}} = 1.16$$

$$Fo = \frac{\alpha t}{L^2} = \frac{4.19 \times 10^{-6} \text{ m}^2/\text{s} \times 180 \text{ s}}{(0.03 \text{ m})^2} = 0.84$$

it follows from Equation 5.44 that

$$\theta_o^* = \frac{\theta_o}{\theta_i} = C_1 \exp(-\zeta_1^2 Fo)$$

where, with $Bi = 0.862$, $C_1 = 1.109$ and $\zeta_1 = 0.814 \text{ rad}$ from Table 5.1. With $Fo = 0.84$,

$$\frac{\theta_o}{\theta_i} = \frac{T(0, 3 \text{ min}) - T_{\infty}}{T_i - T_{\infty}} \bigg|_{\text{Plane wall}} = 1.109 \exp[-(0.814 \text{ rad})^2 \times 0.84] = 0.636$$

Similarly, for the infinite cylinder, with

$$Bi^{-1} = \frac{k}{hr_o} = \frac{17.4 \text{ W/m} \cdot \text{K}}{500 \text{ W/m}^2 \cdot \text{K} \times 0.04 \text{ m}} = 0.87$$

$$Fo = \frac{\alpha t}{r_o^2} = \frac{4.19 \times 10^{-6} \text{ m}^2/\text{s} \times 180 \text{ s}}{(0.04 \text{ m})^2} = 0.47$$

it follows from Equation 5.52c that

$$\theta_o^* = \frac{\theta_o}{\theta_i} = C_1 \exp(-\zeta_1^2 Fo)$$

where, with $Bi = 1.15$, $C_1 = 1.227$ and $\zeta_1 = 1.307 \text{ rad}$ from Table 5.1. With $Fo = 0.47$,

$$\frac{\theta_o}{\theta_i} \bigg|_{\text{Infinite cylinder}} = 1.109 \exp[-(1.307 \text{ rad})^2 \times 0.47] = 0.550$$

Hence, for the center of the cylinder,

$$\frac{T(0, 0, 3 \text{ min}) - T_{\infty}}{T_i - T_{\infty}} = 0.636 \times 0.550 = 0.350$$

$$T(0, 0, 3 \text{ min}) = 300 \text{ K} + 0.350(600 - 300) \text{ K} = 405 \text{ K}$$

The temperature at the center of a circular face may be obtained from the requirement that

$$\frac{T(0, L, 3 \text{ min}) - T_{\infty}}{T_i - T_{\infty}} = \frac{T(L, 3 \text{ min}) - T_{\infty}}{T_i - T_{\infty}} \bigg|_{\text{Plane wall}} \cdot \frac{T(0, 3 \text{ min}) - T_{\infty}}{T_i - T_{\infty}} \bigg|_{\text{Infinite cylinder}}$$

where, from Equation 5.43b,

$$\frac{\theta^*}{\theta_o} = \frac{\theta}{\theta_o} = \cos(\zeta_1 x^*)$$

Hence, with $x^* = 1$, we have

$$\frac{\theta(L)}{\theta_o} = \frac{T(L, 3 \text{ min}) - T_\infty}{T(0, 3 \text{ min}) - T_\infty} \Big|_{\text{Plane wall}} = \cos(0.814 \text{ rad} \times 1) = 0.687$$

Hence

$$\begin{aligned} \frac{T(L, 3 \text{ min}) - T_\infty}{T_i - T_\infty} \Big|_{\text{Plane wall}} &= \frac{T(L, 3 \text{ min}) - T_\infty}{T(0, 3 \text{ min}) - T_\infty} \Big|_{\text{Plane wall}} \cdot \frac{T(0, 3 \text{ min}) - T_\infty}{T_i - T_\infty} \Big|_{\text{Plane wall}} \\ \frac{T(L, 3 \text{ min}) - T_\infty}{T_i - T_\infty} \Big|_{\text{Plane wall}} &= 0.687 \times 0.636 = 0.437 \end{aligned}$$

Hence

$$\frac{T(0, L, 3 \text{ min}) - T_\infty}{T_i - T_\infty} = 0.437 \times 0.550 = 0.240$$

$$T(0, L, 3 \text{ min}) = 300 \text{ K} + 0.24(600 - 300) \text{ K} = 372 \text{ K}$$

The temperature at the midheight of the side may be obtained from the requirement that

$$\frac{T(r_o, 0, 3 \text{ min}) - T_\infty}{T_i - T_\infty} = \frac{T(0, 3 \text{ min}) - T_\infty}{T_i - T_\infty} \Big|_{\text{Plane wall}} \cdot \frac{T(r_o, 3 \text{ min}) - T_\infty}{T_i - T_\infty} \Big|_{\text{Infinite cylinder}}$$

where, from Equation 5.52b,

$$\frac{\theta^*}{\theta_o} = \frac{\theta}{\theta_o} = J_0(\zeta_1 r^*)$$

With $r^* = 1$ and the value of the Bessel function determined from Table B.4,

$$\frac{\theta(r_o)}{\theta_o} = \frac{T(r_o, 3 \text{ min}) - T_\infty}{T(0, 3 \text{ min}) - T_\infty} \Big|_{\text{Infinite cylinder}} = J_0(1.307 \text{ rad} \times 1) = 0.616$$

Hence

$$\frac{T(r_o, 3 \text{ min}) - T_\infty}{T_i - T_\infty} \Big|_{\text{Infinite cylinder}} = \frac{T(r_o, 3 \text{ min}) - T_\infty}{T(0, 3 \text{ min}) - T_\infty} \Big|_{\text{Infinite cylinder}}$$

Hence

$$\frac{T(r_o, 0, 3 \text{ min}) - T_\infty}{T_i - T_\infty} = 0.636 \times 0.339 = 0.216$$

$$T(r_o, 0, 3 \text{ min}) = 300 \text{ K} + 0.216(600 - 300) \text{ K} = 365 \text{ K}$$



Comments:

1. Verify that the temperature at the edge of the cylinder is $T(r_o, L, 3 \text{ min}) = 344 \text{ K}$.
2. The Heisler charts of Section 5S.1 could also be used to obtain the desired results. Accessing these charts, one would obtain $\theta_o/\theta_i|_{\text{Plane wall}} \approx 0.64$, $\theta_o/\theta_i|_{\text{Infinite cylinder}} \approx 0.55$, $\theta(L)/\theta_o|_{\text{Plane wall}} \approx 0.68$, and $\theta(r_o)/\theta_o|_{\text{Infinite cylinder}} \approx 0.61$, which are in good agreement with results obtained from the one-term approximations.
3. The *IHT Models, Transient Conduction* option for the *Plane Wall* and *Infinite Cylinder* may be used to calculate temperature ratios required for the foregoing product solution.

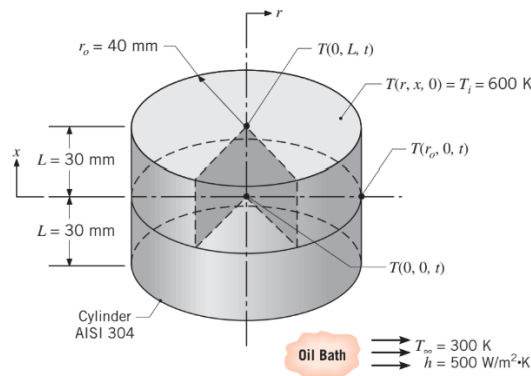
In a manufacturing process stainless steel cylinders (AISI 304) initially at 600 K are quenched by submersion in an oil bath maintained at 300 K with $h = 500 \text{ W/m}^2 \cdot \text{K}$. Each cylinder is of length $2L = 60 \text{ mm}$ and diameter $D = 80 \text{ mm}$. Consider a time 3 min into the cooling process and determine temperatures at the center of the cylinder, at the center of a circular face, and at the midheight of the side.

SOLUTION

Known: Initial temperature and dimensions of cylinder and temperature and convection conditions of an oil bath.

Find: Temperatures $T(r, x, t)$ after 3 min at the cylinder center, $T(0, 0, 3 \text{ min})$, at the center of a circular face, $T(0, L, 3 \text{ min})$, and at the midheight of the side, $T(r_o, 0, 3 \text{ min})$.

Schematic:



Assumptions:

1. Two-dimensional conduction in r and x .
2. Constant properties.

Properties: Table A.1, stainless steel, AISI 304 [$T = (600 + 300)/2 = 450 \text{ K}$]: $\rho = 7900 \text{ kg/m}^3$, $c = 526 \text{ J/kg} \cdot \text{K}$, $k = 17.4 \text{ W/m} \cdot \text{K}$, $\alpha = k/\rho c = 4.19 \times 10^{-6} \text{ m}^2/\text{s}$.

Analysis: The solid steel cylinder corresponds to case (i) of Figure 5S.11, and the temperature at any point in the cylinder may be expressed as the following product of one-dimensional solutions.

$$\frac{T(r, x, t) - T_\infty}{T_i - T_\infty} = P(x, t)C(r, t)$$

Appendix B - Heat Maps

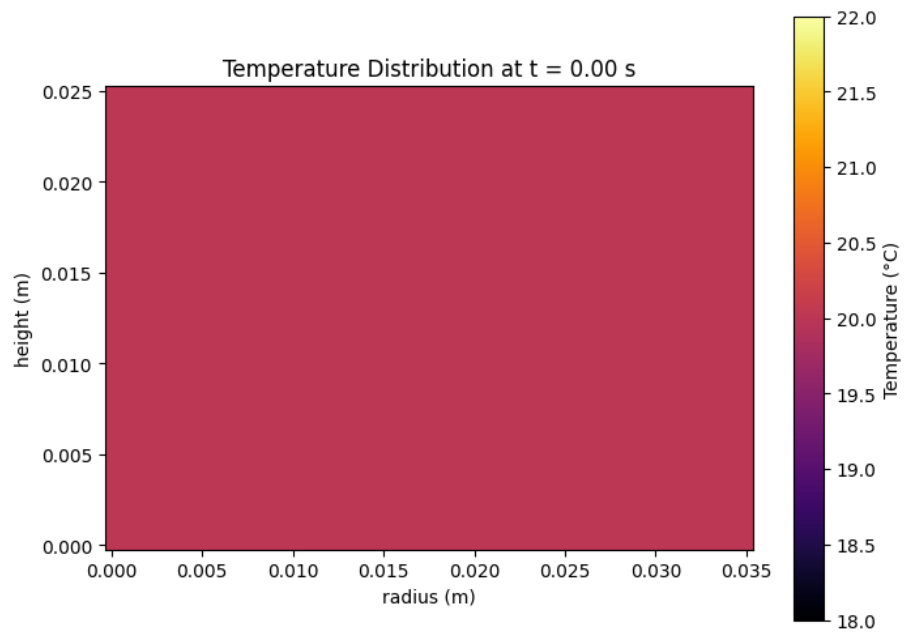


Figure 6: Time: 0 s

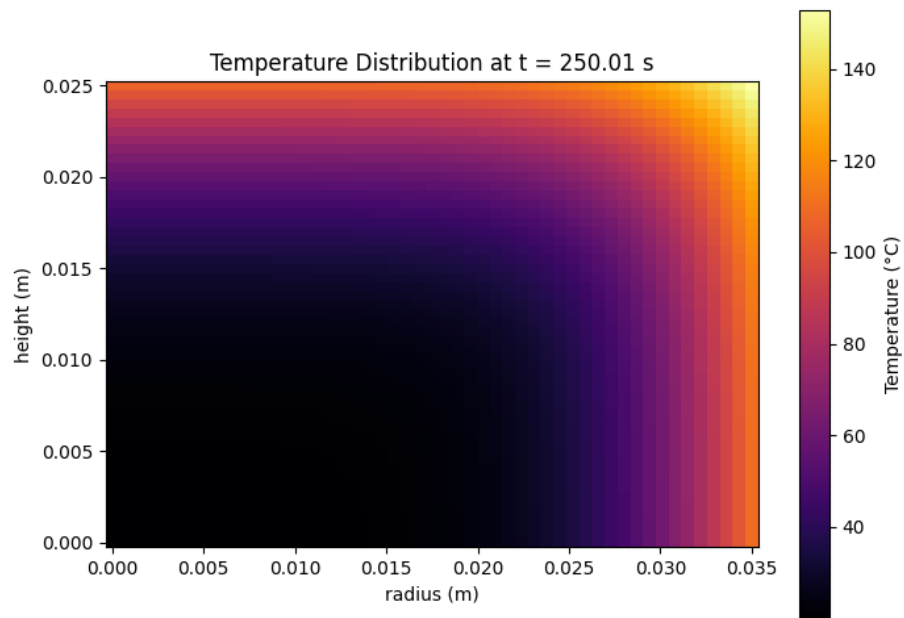


Figure 7: Time: 250 s

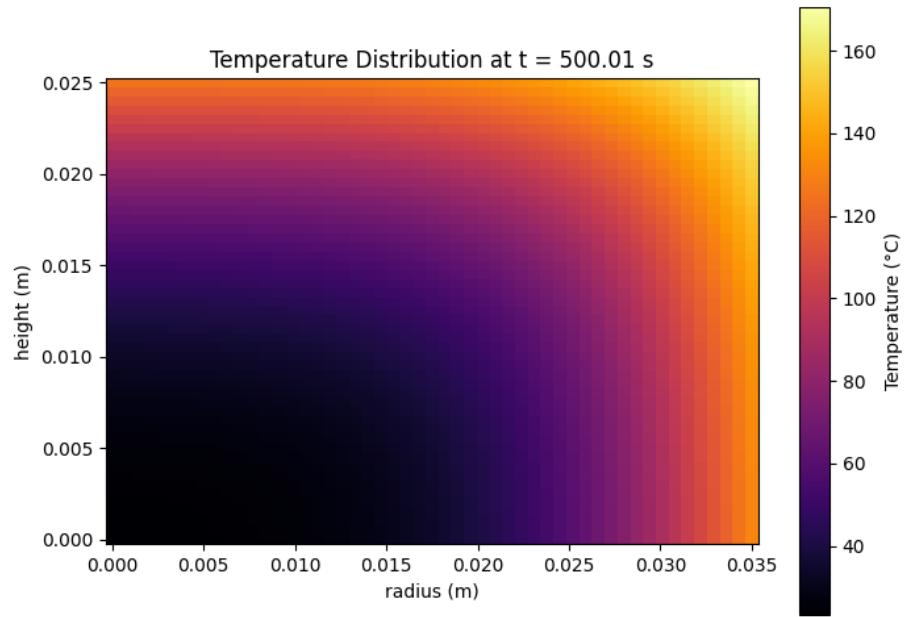


Figure 8: Time: 500 s

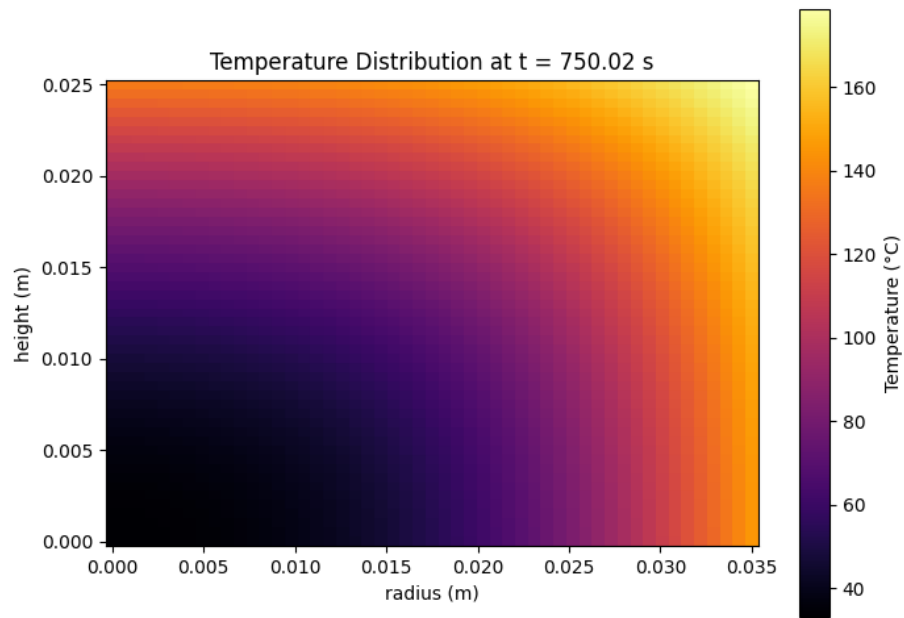


Figure 9: Time: 750 s

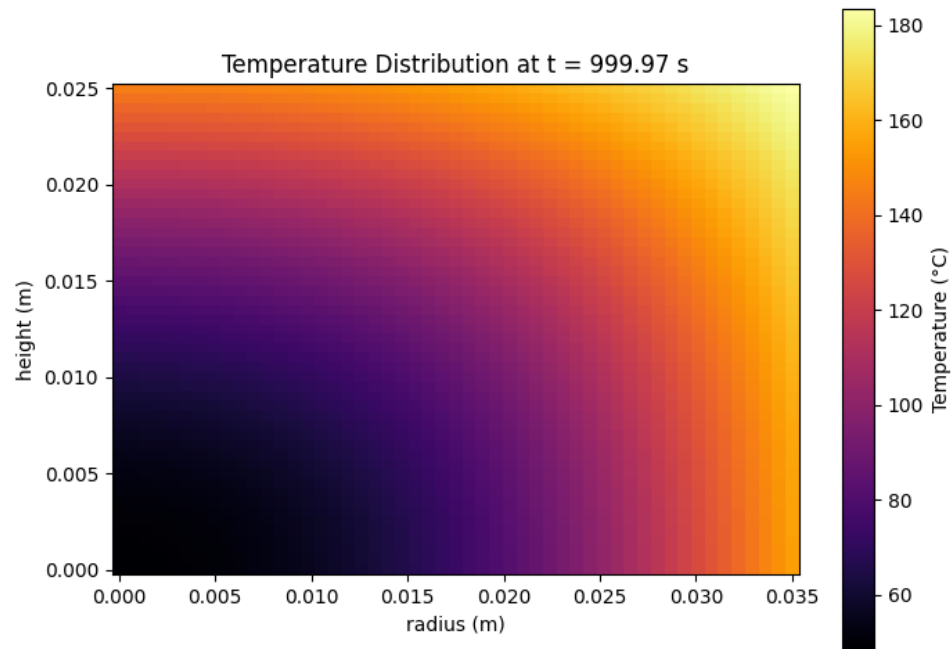


Figure 10: Time: 1000 s

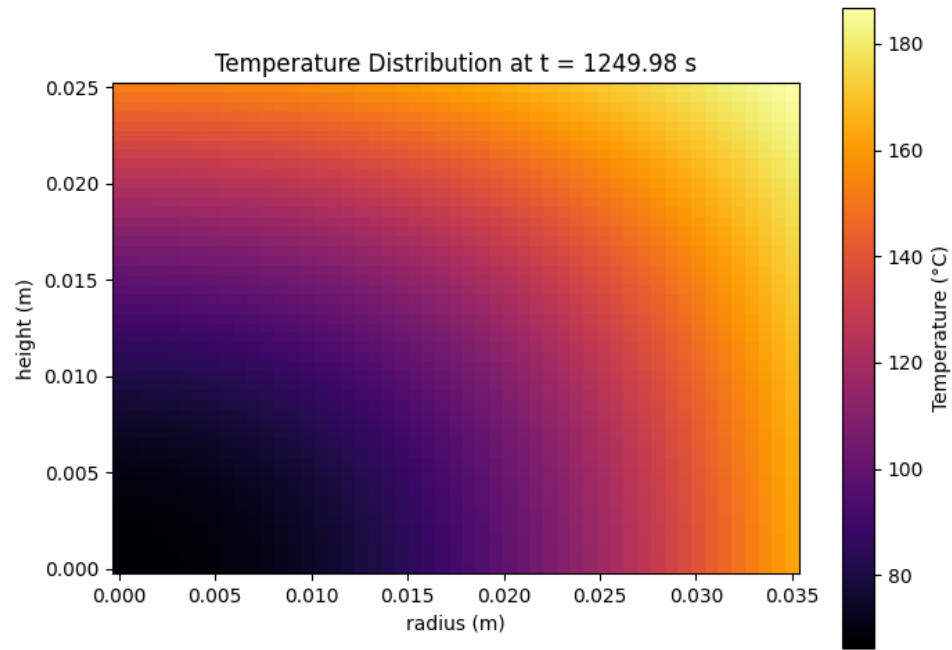


Figure 11: Time: 1250 s

References